We want to help farmers:

1) if we want the agents to help them, they need

1. The land

- **Size**: how many hectares (1 hectare = about a football field).
- **Soil type**: sandy (water drains fast), clay (holds water, heavy), or in between. This decides what grows well.
- **Slope and location**: water runs off steep land, and some areas get frost or strong wind.

2. Water

- **Source**: well, river, dam, canal, rain only.
- **How much** he can use per season, and **what it costs** (pumping, electricity, fuel).
- **Quality**: salty water harms many plants.
- **Method**: flooding the field (wasteful), sprinklers, or drip (small tubes dripping at each plant, which saves a lot of water).

3. Weather

- Rain, heat, cold, wind, drought risk.
- Each plant has its own tolerance. Avocado needs lots of water, olives and cactus need little, and some plants break in strong wind.

4. What to plant

- **Seeds** (wheat, vegetables) or **young plants/trees** (olives, citrus, avocado).
- Trees take **years** before giving fruit, while vegetables give results in **months**. This matters a lot for money.

5. Feeding the plants

- **Fertilizer**: plant food. Some is chemical, some is natural (compost, manure).
- The right amount at the right time. Too much wastes money and hurts the soil.

6. Protection

- **Pests and diseases**: insects, fungus.
- **Weeds**: they steal water and food.
- **Wind, frost, hail**: nets, windbreaks (rows of trees), covers.

7. Work and equipment

- **People** to plant, care for and harvest (often the biggest cost at harvest time).
- **Tools and machines**: tractor, pump, pipes, harvesting tools.
- **Energy**: electricity or fuel for pumps.

8. Timing (the calendar)

- When to prepare the soil, plant, water, feed, and harvest. Planting two weeks too late can ruin a season.

9. After the harvest

- **Storage**: cold rooms or dry storage, so food doesn't rot.
- **Transport** to the market.
- **Packaging and sorting** for better prices.

#	Agent	Covers your topics	What it does
1	**Field Data Agent**	1. Land, 3. Weather	Fetches soil, slope, location, and weather (rain, heat, cold, wind, drought risk) from APIs. Outputs a "profile" of the parcel.
2	**Water Agent**	2. Water	Computes available water, cost per m^3, quality check, and the best method (drip vs sprinkler vs flooding).
3	**Crop Selection Agent**	4. What to plant	Picks viable crops (seeds vs trees) and rejects impossible ones (e.g. avocado in drought). Run **3 versions** with different goals: save water, maximize profit, minimize risk.
4	**Crop Care Agent**	5. Feeding, 6. Protection	Fertilizer plan, pest and weed control, wind/frost/hail protection (nets, windbreaks, covers).
5	**Resources Agent**	7. Work &	Estimates labor,

		equipment	machines, and energy needs and costs for the plan.
6	**Calendar Agent**	8. Timing	Builds the full timeline (prepare, plant, water, feed, harvest) and adjusts it to weather forecasts.
7	**Post-Harvest Agent**	9. After harvest	Plans storage, transport, packaging, and sorting to protect quality and price.
8	**Judge / Orchestrator Agent**	All	Collects everything, compares the 3 strategies, picks the best compromise, explains it simply, and asks for improvements if a plan is weak.

This is also your optimization map for later: agents 1, 2 and 5 are not “real” LLM agents, so they’re candidates to present as tools rather than agents.